

Does a Slime Mold Think?

What counts as cognition — and does biology matter?

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drafted with Claude (Anthropic)

Computational Modelling for Psychology & Neuroscience

Where we've been

- ▶ **Dog · bell · food** — nobody argues. We grant “it learned.”
- ▶ **Sunflower · sun · sugar** — anticipation of dawn, but *built in*. Not learned.
- ▶ **Pea plant · fan · light** — *claimed* associative learning. Contested; a replication failed.

A single cell with no brain

Physarum polycephalum — a slime mold.

- ▶ One giant **multinucleate cell**. No neurons, no synapses, no nervous system.
- ▶ Forages by growing a living network of tubes; routes are reinforced or pruned by internal **flow feedback**.
- ▶ Long filed under “fungus”; actually an amoeba.

Everything on the next slides is done by *this*.

It solves mazes

Place it between two food sources in a maze. It fills the maze, then **withdraws from dead ends** and keeps the tube along the **shortest path** connecting the food.

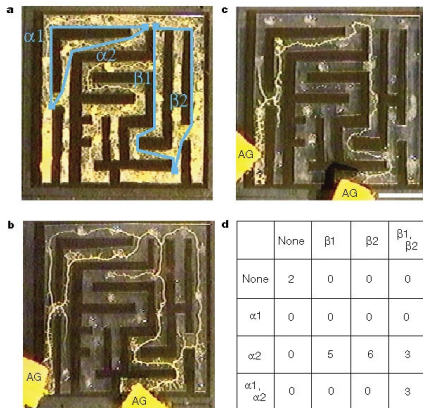


Figure 1: Maze Navigation in the Slime Mold

It designs networks that rival engineers

Arrange oat flakes as the cities of Greater Tokyo. The mold grows a network that **approximates the real rail system** — balancing total tube length, transport efficiency, and **fault tolerance** against accidental breaks.

Those are the exact tradeoffs human planners spend months negotiating.

— *Tero et al. (2010), Science*

But is it *learning*?

The sunflower failed this test. The slime mold does not.

- ▶ **Habituation** — made to cross a bridge laced with a *harmless* bitter substance, it hesitates at first, then stops reacting over days — and recovers if left alone. Habituation is a textbook elementary form of learning.
- ▶ **Anticipation** — shocked with cold at fixed intervals, it slows *before* the next shock — and still slows once at the expected time when the shock is withheld.

— *Boisseau, Vogel & Dussutour (2016); Saigusa et al. (2008)*

Memory — without a brain ...

and outside the body

It avoids ground it has already covered by **sensing the extracellular slime it left behind**. The record of where it has been is written into the *environment*, not stored in a head.

— *Reid, Latty, Dussutour & Beaton (2012), PNAS*

So: Does it Think?

Which definition are you using?

- ▶ **Functional** — cognition is what a system *does* (solve, learn, remember). → It qualifies.
- ▶ **Substrate** — cognition is a property of certain *stuff* / *architecture* (nervous systems). → It doesn't.

Is a mathematical model of the slime mold computational cognitive (neuro)science?

A mathematical model of slime mold shortest path computation.

Math Model

Can an organism *be* an algorithm?

To answer “does the slime mold think?” do you have to decide if **the computation it runs is cognition?**

— *Tero et al. (2010); Bonifaci, Mehlhorn & Varma (2012)*

Our four test cases compared

“X learns that Y predicts Z” hides four separate claims. Line the cases up against them:

Does the case have. . .	Dog	Sunflower	Pea	Slime mold	LLM
a real $Y \rightarrow Z$	yes	yes	yes	yes	yes
contin- gency behaviour that tracks it	yes	yes	<i>claimed</i>	yes	yes
acquisition by experience	yes	no (built-in)	<i>claimed</i>	yes	train: yes / run: no
an internal representa- tion	yes?	a clock?	<i>claimed</i>	external?	<i>contested</i>

- ▶ The **top two rows are trivial**. A thermostat passes.
- ▶ **Acquisition** is *learning*. Is the sunflower learning? No. The dog and slime mold? Yes.
- ▶ **Representation** — this is where we assume people live and where cognition is defined. But then don't we have to grant the pea plant and slime mold cognitive status?gs, one word.)*

The Assignment

Write your answer to the question,

“Does a slime mold think?”

One page maximum. As long as I can read it you are good. No worries about spelling or grammar, but don't be careless. I am not looking for a particular answer. I will be reading for well reasoned responses.

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